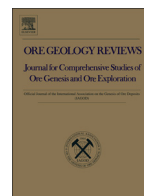




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Machine learning predictive models for mineral prospectivity: An evaluation of neural networks, random forest, regression trees and support vector machines

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ABSTRACT

Machine learning algorithms (MLAs) such as artificial neural networks (ANNs), regression trees (RTs), random forest (RF) and support vector machines (SVMs) are powerful data driven methods that are relatively less widely used in the mapping of mineral prospectivity, and thus have not been comparatively evaluated together thoroughly in this field.

The performances of a series of MLAs, namely, artificial neural networks (ANNs), regression trees (RTs), random forest (RF) and support vector machines (SVMs) in mineral prospectivity modelling are compared based on the following criteria: i) the accuracy in the delineation of prospective areas; ii) the sensitivity to the estimation of hyper-parameters; iii) the sensitivity to the size of training data; and iv) the interpretability of model parameters. The results of applying the above algorithms to epithermal Au prospectivity mapping of the Rodalquilar district, Spain, indicate that the RF outperformed the other MLA algorithms (ANNs, RTs and SVMs). The RF algorithm showed higher stability and robustness with varying training parameters and better success rates and ROC analysis results. On the other hand, all MLA algorithms can be used when ore deposit evidences are scarce. Moreover the model parameters of RF and RT can be interpreted to gain insights into the geological controls of mineralization.

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1. Introduction

The development or application of a transparent and reproducible approach for identifying locations with a high potential to be explored further is a main goal for a study on mineral prospectivity (Joly et al., 2012). The most critical procedure in prospectivity modelling is the selection of appropriate targeting criteria and the application of innovative and robust techniques for derivation of the evidential features for these criteria (Joly et al., 2012). However, the methodological aspects are also important. The analysis of the spatial relationships between evidential features and known deposit locations is carried out by means of different numerical methods (Bonham-Carter, 1994; Carranza, 2008). Hence, selecting a suitable methodology or algorithm is essential to obtain an accurate mineral potential map. It depends, mainly, on the capacity of the algorithm to learn complex relationships between the input evidential features and the occurrence of mineral deposits, but

also interpretability and transparency must be considered. However, in most practical applications, the algorithm is selected based on ease of implementation and availability of software tools. Hence, it is necessary to investigate new robust methods which are transparent and operative at the same time.

In the past few decades numerous methods have been applied which can be grouped into two sets: knowledge-driven models and data-driven models. The parameters of knowledge-driven models are estimated based on the expert knowledge of the processes that resulted in the formation of mineral deposits in the given geological setting (Abedi et al., 2013; Carranza, 2008). On the other hand, the parameters of data-driven models are estimated based on quantitative measures of spatial associations between evidential features and known deposit locations (Carranza, 2011). The numerical models traditionally used in mineral prospectivity mapping (data-driven models) are probabilistic models such as discriminant analysis (Chung, 1977; Harris et al., 2003) or logistic regression (Chen et al., 2011; Chung, 1978; Fallon et al., 2010; Mejía-Herrera et al., 2014; Porwal et al., 2010a) and a set of methods known as artificial intelligence or machine learning (Lewkowski et al., 2010; Oh and Lee, 2010; Pereira Leite and de Souza

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Filho, 2009a,b; Porwal et al., 2003, 2010b; Rigol-Sanchez et al., 2003; Singer and Kouda, 1996), among others. For a detailed review, please refer to Carranza (2011). Several studies demonstrate that this last group, machine learning algorithms (MLAs), are more accurate than statistical techniques such as discriminant analysis or logistic regression, especially when the feature space is complex (i.e. when the dimensionality of the input feature space is expected to be high and the relationship between the targeted deposits and the input evidential feature is expected to be non-linear) or the input datasets are expected to have different statistical distributions (Abedi et al., 2012; Brown et al., 2000; Harris et al., 2003; Piccini et al., 2012; Zuo and Carranza, 2011). MLAs have the potential to identify and model the complex non-linear relationships between the mineral occurrences and the evidential features (Brown et al., 2000). These methods can handle a large number of evidential features which might be important in mineral prospectivity studies. However, increasing the number of input evidential features may lead to increased complexity and larger numbers of model parameters, thus the model becomes susceptible to over fitting because of the curse of dimensionality (Bellman, 2003; Rodriguez-Galiano et al., 2012a).

In the past few decades a large number of methods for classification have been developed (Hastie et al., 2009). Among the most widely used techniques are decision trees (DTs) (Breiman et al., 1984), artificial neural networks (ANNs) (Brown et al., 2000; Porwal et al., 2003; Rigol-Sanchez et al., 2003; Rumelhart et al., 1986), support vector machines (SVMs) (Abedi et al., 2012; Boser et al., 1992; Cortes and Vapnik, 1995; Zuo and Carranza, 2011) and ensembles of classification trees such as random forest (RF) (Breiman, 2001; Rodriguez-Galiano et al., 2014). Two ANN algorithms are already implemented in operational GIS applications for mineral prospectivity (Avantra Geosystems, 2006; Kemp et al., 1999; Sawatzky et al., 2009, 2010), which explains why these are the most widely used MLAs in mineral prospectivity modelling. It remains nevertheless to be questioned whether ANN algorithms are the best tools for mineral potential mapping, gaining insights in modelling retrieval performances. Besides, training ANNs require the estimation of values for numerous parameters that may greatly impact the final robustness of the model. Algorithms based on DT are easy to apply, as fewer number of parameters need be estimated; hence, these have high degrees of automation (Bater and Coops, 2009; Herrera et al., 2010). However, this comparative advantage of DT with respect to ANN can be hidden by a tendency to over fit data (Breiman, 1984). For these reasons, both ANN and DT are being replaced by a more advanced, simpler to train MLA in recent years. During the past decade, the family of kernel methods such as SVM (Al-Anazi and Gates, 2010; Booker and Snelder, 2012; Chen et al., 2013; Zhao et al., 2012; Zimmermann et al., 2012) and ensembles of trees such as RF (Chan and Paelinckx, 2008; Davis and Robinson, 2012; Ghimire et al., 2012; Rodriguez-Galiano et al., 2012b; Vincenzi et al., 2011; Wang et al., 2009; Waske and Braun, 2009) have emerged as very promising methodologies for geosciences. However, those studies using MLA for mineral prospectivity are limited, especially in the case of the newest algorithms such as RF. In the case of SVMs, their parameterisation needs or operativity have not been studied in depth (Abedi et al., 2012; Zuo and Carranza, 2011). Moreover, most studies have not attempted to understand the performance of the machine learning algorithms using scarce training data.

The aim of this study is to test the capabilities of four machine learning regression algorithms (ANN, DT, RF and SVM) for predictive modelling of epithermal gold potential from geological, geochemical, geophysical and EO-1 Hyperion derived information. These algorithms were specifically chosen as although they are being increasingly used in Earth and environmental sciences, yet have not been compared with one another exhaustively in mineral prospectivity modelling. The comparative analysis carried out was approached from different perspectives: the mapping accuracy, parameterisation needs of each method and sensitivity to the training sample size, as well as the interpretability of model parameters. Thus the following questions are investigated in this paper: i) Are ANN, RF, RT and SVM equally accurate in the delineation of prospective areas

for mineral deposits of the type sought?; ii) Are the predictions of these methods over-sensitive to their hyper-parameters? – or, in other words, which method is the easiest to apply; iii) Can these algorithms be applied in situations in which the number of known deposit locations is scarce?; iv) Which method offers more information about the relationship between epithermal Au occurrences and evidential features?

MLAs were applied to a comprehensive exploration database for mineral potential mapping in the Rodalquilar gold mining district (Spain). This district is a favourable area in order to carry out pilot studies given the abundant information and the previous published works that make it a reasonable database for comparison of results and robustness of the methodology (Rodriguez-Galiano et al., 2014). Several studies have also been published using remote sensing for geological or mineral potential mapping in this district. Rigol and Chica-Olmo (1998) applied image fusion techniques for geological–environmental mapping. Chica-Olmo et al. (2002) developed a mineral exploration decision support system for gold potential mapping in the Rodalquilar–San Jose districts. Rigol-Sanchez et al. (2003) proposed an artificial neural network model for gold prospectivity mapping in the Rodalquilar district. van der Meer (2006) and Bedini et al. (2008) used HyMap imaging spectrometer data to map mineralogy in the Rodalquilar caldera. Carranza et al. (2008) proposed a new hybrid model based on evidential belief functions. Debba et al. (2009) developed a new methodology to derive optimal exploration target zones in the Rodalquilar district. Moreover, there are several studies aimed at evaluating the environmental impact of mining activities in the area using remote sensing data (Choe et al., 2008; Ferrier et al., 2007, 2009) or geochemistry (Bagur et al., 2009; Flores and Rubio, 2010; Oyarzun et al., 2009). It is worth mentioning that, from the standpoint of remote sensing, the use of EO1-Hyperion images in this paper is innovative with respect to previous papers in which AVIRIS (Ferrier and Wadge, 1996), Landsat-5 TM (Crosta and Moore, 1989; Rigol-Sanchez et al., 2003; Rodriguez-Galiano et al., 2014), ASTER (Carranza et al., 2008), or HyMap (Bedini et al., 2008; Ferrier et al., 2007; van der Meer, 2006) images were used.

2. Machine learning algorithms

2.1. Artificial neural networks

The most common approach to develop nonparametric and nonlinear classification/regression is based on ANNs. There are many different types of ANNs. However, it is not the scope of this paper to describe the different types of networks, which can be found at the bibliography. This section provides a brief description of one of the most used ANNs: the feed-forward propagation neural network (Rumelhart et al., 1986).

As in the brain, the basic processing elements of an artificial neural network are neurons (units or nodes). In a neural network, units are placed as layers, and are connected in such a way that information flows unidirectionally, from the input units – through the unit or units located on the hidden layer/layers – to the units on the output layer. Input units distribute the signal to the hidden units of the second layer. A neuron basically performs a linear regression followed by a non-linear function, $f(\cdot)$. Neurons of different layers are interconnected with the corresponding links (weights). In this paper, we have used the standard multi-layer ANN model, whose neuron j in layer $l + 1$ yields $x_j^{l+1} = f(\sum_i w_{ij}^l x_i^l + w_{bj}^l)$, where w_{ij}^l are the weights connecting neuron i in layer l to neuron j in layer $l + 1$, w_{bj}^l is the bias term of neuron j in layer l , and f is a logistic activation function. The prediction of the model for the sample x_i is denoted as $f(x_i)$. The aim of the algorithm is to find a set of weights which ensures that, for each input vector, the resulting vector from the network is the same, or close enough, to the desired output vector. If there is a definite and finite set of input-output cases (patterns), the overall error in the functioning of the network with a particular set of weights can be calculated by comparing the real and desired output vectors for each pattern, for example, by

the method of least squares. Training an ANN needs selecting a structure (number of hidden layers and nodes per layer), the proper initialisation of the weights, learning rate, and regularisation parameters to prevent over fitting.

2.2. Regression trees

DTs, along with neural networks, are the most widely used machine learning algorithms in geosciences (Friedl and Brodley, 1997; Hansen et al., 1996; Lippitt et al., 2008; Pal and Mather, 2003; Rogan et al., 2003; Wessels et al., 2004). The increasing use of DT is linked to their simplicity and interpretability, their low computational cost and to the possibility of being graphically represented. A DT represents a set of restrictions or conditions which are hierarchically organized, and which are successively applied from a root to a terminal node or leaf of the tree (Breiman, 1984; Quinlan, 1993). The main benefit of using a hierarchical tree structure to perform classification decisions is that the tree structure is transparent, which in comparison with artificial neural networks (ANNs), is easier to interpret. In order to induce the DT from a dataset, an evaluation measure of each of the evidential features is used to maximise the inter node heterogeneity.

Two different methodologies can be distinguished within DT: classification trees and regression trees (RT). This section presents a brief review of the theoretical basis of RT, considered more suitable for the intended purpose. In order to induce the DT, recursive partitioning and multiple regressions are carried out from the dataset. From the root node, the data splitting process in each internal node of a rule of the tree is repeated until a stop condition previously specified is reached. Each of the terminal nodes, or leaves, has attached to it a simple regression model which applies in that node only. Once the tree's induction process is finished, pruning can be applied with the aim of improving the tree's generalisation capacity by reducing its structural complexity. The number of cases in nodes can be taken as pruning criteria.

As described by Breiman et al. (1984) the induction of the DT involves first selecting optimal splitting measurement vectors. The process starts by splitting the dependent feature, or the parent node (root), into binary pieces, where the child nodes are 'purer' than the parent node. Through this process, the DTs search through all candidate splits to find the optimal split, s^* , that maximises the 'purity' of the resulting tree (as defined by the largest decrease in the impurity).

$$\Delta i(s, t) = i(t) - p_L i(t_L) - p_R i(t_R)$$

In this equation, s is the candidate split at node t , and the node t is divided by s into the left child node t_L with a proportion of p_L , and right child node t_R with a proportion of p_R . $i(t)$ is a measure of impurity before splitting, $i(t_L)$ and $i(t_R)$ are measures of impurity after splitting, and $\Delta i(s, t)$ measures the decrease in impurity from split s .

There are many approximations for measuring impurity. Some of the most frequent ones are gain-ratio (Quinlan, 1993), Gini index (Breiman et al., 1984) and Chi-square (Mingers, 1989). The most common measure is the Gini index. The Gini index used in this research measures $i(t)$ as the

$$I_G(t_{X(x_i)}) = 1 - \sum_{j=1}^m f(t_{X(x_i)}, j)^2$$

where $f(t_{X(x_i)}, j)$ is the proportion of samples with the value x_i belonging to leave j as node t . The decision tree splitting criterion is based on choosing the attribute with the lowest Gini impurity index (I_G).

2.3. Random forest

RF is a regression technique that combines the performance of numerous DT algorithms to classify or predict the value of a variable (Breiman, 2001; Guo et al., 2011; Rodríguez-Galiano et al., 2012b). That is when RF receives an (x) input vector, made up of the values of the different evidential features analysed for a given training area, RF builds a number K of regression trees and averages the results. After K such trees $\{T(x)\}_1^K$ are grown, the RF regression predictor is:

$$\hat{f}_{rf}^K(x) = \frac{1}{K} \sum_{k=1}^K T(x).$$

To avoid the correlation of the different trees, RF increases the diversity of the trees by making them grow from different training data subsets created through a procedure called bagging. Bagging is a technique used for training data creation by resampling randomly the original dataset with replacement, i.e., with no deletion of the data selected from the input sample for generating the next subset $\{h(x, \Theta_k), k = 1, \dots, K\}$, where $\{\Theta_k\}$ are independent random vectors with the same distribution. Hence, some data may be used more than once in the training, while others might never be used. Thus, greater stability is achieved, as it makes it more robust when facing slight variations in input data and, at the same time, it increases prediction accuracy (Breiman, 2001). On the other hand, when the RF makes a tree grow, it uses the best feature/split point within a subset of evidential features which has been selected randomly from the overall set of input evidential features. Therefore, this can decrease the strength of every single tree, but it reduces the correlation between the trees, which reduces the generalisation error (Breiman, 2001). Another characteristic of interest is that the trees of a RF classifier grow with no pruning, which makes them light, from a computational perspective.

Additionally, the samples which are not selected for the training of the k -th tree in the bagging process are included as part of another subset called out-of-bag (oob). These oob elements can be used by the k -th tree to evaluate performance (Peters et al., 2007). In this way RF can compute an unbiased estimation of the generalisation error without using an external test data subset (Breiman, 2001). The generalisation error converges as the number of trees increases; therefore, the RF does not over fit the data. RF also provides an assessment of the relative importance of the different evidential features. This aspect is useful for multi-source studies, where data dimensionality is very high, and it is important to know how each feature influences the prediction model to be able to select the best evidential features (Gislason et al., 2006; Pal, 2005). To assess the importance of each variable (e.g. satellite image band), the RF switches one of the input evidential features while keeping the rest constant, and it measures the decrease in accuracy which has taken place by means of the oob error estimation (Breiman, 2001).

2.4. Support vector machines

Although SVMs were proposed by Vapnik in the late 1960s, they have not received significant attention until recent years when they have become a promising estimator in data-driven fields. SVM is a supervised method to perform dichotomy classification of multidimensional feature-vectors (Vapnik and Chervonenkis, 1964; Vapnik and Lerner, 1963). Originally, it was developed as a linear classification method, generalised later to a non-linear classifier and, lastly, it was extended to regression problems (Cortes and Vapnik, 1995).

The basic idea under the SVM method is to transform the input features into a higher-dimensional space where the two classes can be linearly separated by a high-dimensional surface, known as hyper-plane. Given a training dataset $\{\mathbf{x}_i\}_{i=1}^N$ with N samples, where $\mathbf{x} \in \mathbb{R}^L$ is a vector of L input-features, and its corresponding known output-features

$\{y_n\}_{n=1}^N = 1$, with $y_n \in \{-1, 1\}$, the SVM regression model is defined then as:

$$f(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b$$

where $\phi: \mathbf{x} \rightarrow \phi(\mathbf{x}) \in \mathbb{R}^H$ is any non-linear function that maps the input data into the high-dimensional feature space with $H \geq L$. Originally, assuming linearly separable features, this function was trivially defined as $\phi(\mathbf{x}) = \mathbf{x}$. On the other hand, the unknown parameters of the model are $\mathbf{w}$, a weight vector which is normal to the hyper-plane and b , the hyper-plane bias.

The SVM model for regression is defined then to cope with non-separable features by allowing misclassification errors. Therefore, the SVM model presented above is subject to the following constraints:

$$\begin{aligned} y_n - f(\mathbf{x}_n) &\leq \xi_n + \varepsilon \\ f(\mathbf{x}_n) - y_n &\leq \xi_n + \varepsilon \\ \varepsilon, \xi_n, \xi_n^* &\geq 0, \forall n \end{aligned}$$

where ε is the (in)sensitivity, i.e. the maximum misclassification error allowed and $\{\xi_n, \xi_n^*\}_{n=1}^N$ are slack variables quantifying the output-features deviation from the positive and negative classes.

The optimisation of the previous model, subject to the soft-margin constrain, defines a hyper-plane which separates the training data with the maximum margin. The optimisation problem can be solved by using the Lagrange multipliers method, (for details see Vapnik, 2000), yielding to the next cost function:

$$\begin{aligned} L(\{a_n, a_n^*\}_{n=1}^N) \\ = -\frac{1}{2} \sum_{i,j=1}^N (a_i - a_i^*)(a_j - a_j^*) K(\mathbf{x}_i, \mathbf{x}_j) - \varepsilon \sum_{i=1}^N (a_i + a_i^*) + \sum_{i=1}^N (a_i - a_i^*) y_i \end{aligned}$$

where $\{a_n, a_n^*\}_{n=1}^N$ are the Lagrange multipliers and $K(\mathbf{x}_i, \mathbf{x}_j)$ is the Kernel function, defined as the inner product of the transformed input-feature vectors:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \langle \phi(\mathbf{x}_i) | \phi(\mathbf{x}_j) \rangle.$$

The optimisation of this cost function is significantly simplified by introducing the kernel notation. Instead of designing a mapping function, then transform the data and later compute the inner products, the SVM approach directly defines the kernel as a function of the input-feature vector. Some kernel functions typically considered on SVM applications are shown below:

$$K_{\text{linear}}(\mathbf{x}, \mathbf{x}') = \mathbf{x} \cdot \mathbf{x}'$$

$$K_{\text{polynomial}} = (\gamma \mathbf{x} \mathbf{x}' + r)^p$$

$$K_{\text{RBF}}(\mathbf{x}, \mathbf{x}') = \exp(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2)$$

$$K_{\text{sigmoid}}(\mathbf{x}, \mathbf{x}') = \tanh(\gamma \mathbf{x} \mathbf{x}' + r).$$

Once we estimate $\{\hat{a}_n, \hat{a}_n^*\}_{n=1}^N$ by maximising the cost function defined above, the margin can be inferred as:

$$\hat{\mathbf{w}} = \sum_{n=1}^N (\hat{a}_n - \hat{a}_n^*) \phi(\mathbf{x}_n)$$

such as $f(\mathbf{x})$ can be directly estimated as:

$$\hat{f}(\mathbf{x}) = \sum_{n=1}^N (\hat{a}_n - \hat{a}_n^*) K(\mathbf{x}_i, \mathbf{x}) + \hat{b}$$

where the computation of $\hat{b}$ can be conveniently dropped out by preprocessing and centralising the data, forcing the bias to be zero.

3. Study area

The study area corresponds to the Rodalquilar mining district, which is located in the southeast of Spain, within the province of Almería. Rodalquilar was chosen for this pilot study to test the application of different data driven machine learning methods to mineral potential mapping because it contains a sufficiently large number of gold occurrences to provide training data for the application of this methodology. The Rodalquilar epithermal gold-alunite deposit occurs within the Rodalquilar caldera complex. It is the first documented example of caldera-related epithermal Au mineralisation in Europe (Arribas et al., 1995). This mining district covers an area of 150 km² (Fig. 1) and mostly coincides with the Miocene Cabo de Gata volcanic field, which makes up a mountain range of the same name and goes along the coast from the Cabo de Gata. This area is characterised by epithermal quartz-alunite gold deposits which are associated with felsic to intermediate tertiary volcanic rocks showing fracturing and pervasive hydrothermal alteration (Demoustier et al., 1999; Rytuba et al., 1990). Volcanic rocks range in composition from pyroxene andesite to rhyolite and in age from about 15 to 7 (million years) (Arribas et al., 1995; Zeck et al., 2000). The geodynamic environment of formation of these rocks is controversial. Subduction models (López Ruiz and Rodríguez-Badiola, 1980) or crustal thinning due to postcollisional extensional collapse (Doblas and Oyarzun, 1989) has been proposed. Recent geochemical and geochronological data support an origin of the Alboran Basin through subduction and roll-back of oceanic lithosphere (Duggen et al., 2004).

A brief description of the main aspects related to the mineralization and alteration zones is given below (see Arribas et al. (1995) and Rytuba et al. (1990) for more details).

Mineralisation within the Rodalquilar caldera complex consists of low-sulphidation Pb–Zn–(Cu–Ag–Au) quartz veins and the economically most important high-sulphidation Au–alunite–(Cu–Te–Sn) epithermal deposits. The Au–(Cu–Te–Sn) ores are preferentially localised in ring and radial faults and fractures along the east wall of the Lomilla caldera inside the Rodalquilar caldera. The primary Au mineralisation occurs chiefly as chalcedonic quartz veins and as hydrothermal breccias with high Te and Sn contents. The Au mineralisation is restricted to zones of intensely altered rock, particularly zones of silicic and advanced argillic alteration. The mineralisations are principally related to fractures within the margins of calderas, as well as to regional structures, north–south mainly, through which the mineralising hydrothermal fluids preferentially circulated, and around which zoning of hydrothermal alterations of the wall-rock occurred.

Different alteration types can be distinguished: propylitic, sericitic, intermediate argillic, advanced argillic and silicic (terminology according to Heald et al. (1987)). However, economic Au mineralisation greater than 1 g/t is only found in patches of leached and silicified rock. The silicic zone includes vuggy residual silica and massive silicified rock within halos of advanced argillically altered rocks. The advanced argillic zone is composed mainly of quartz + alunite ± kaolinite – dickite. Other minerals present in this zone include pyrite, pyrophyllite, and illite. These alterations resulted from the reaction of volcanic rocks and extremely acidified fluids. These fluids contained sulphur from a dioritic magma in depth and, very likely, from the sea (Demoustier et al., 1999). It is also believed that the influence of both meteoric and seawaters was key to the precipitation of gold compounds (Arribas et al., 1995). The wall-rocks are mainly tuff, ignimbrites, collapse breccia and rhyolite domes from the Rodalquilar and La Lomilla calderas. In the zones closest to fractures, where there is a maximum alteration and the rock is totally leached, a vuggy-silica alteration takes place made up of vuggy silica surrounded by an advanced argillic alteration, with quartz + kaolinite +

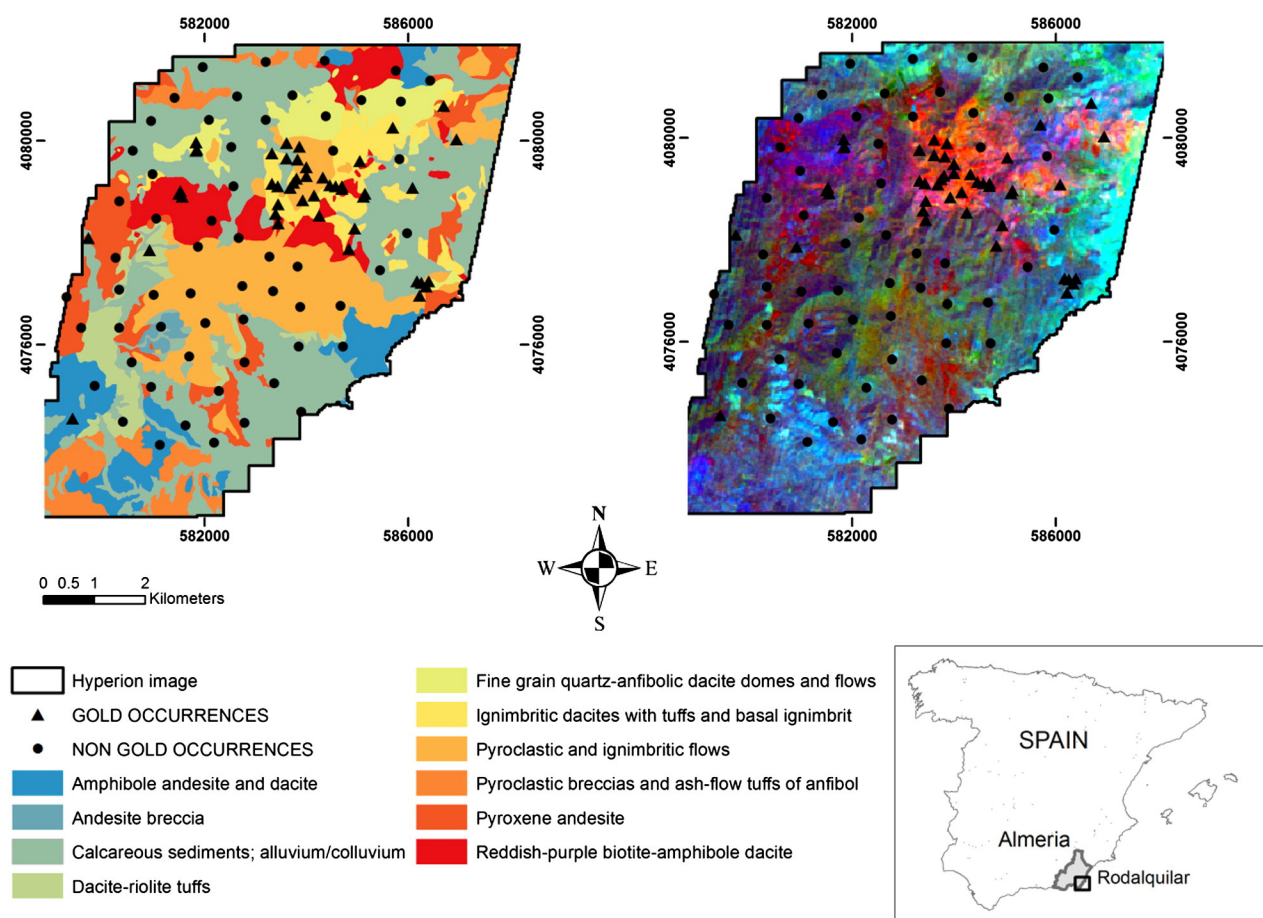


Fig. 1. Location of the study area (bottom right), the distribution of Neogene volcanic rocks and locations of epithermal deposits (left panel) and false colour composition of the main MTMF components derived from the hyper-spectral satellite image Hyperion. Map coordinates are in metres (UTM project, zone 30 N, International 1924 ellipsoid, European 1950 datum).

pyrophyllite + illite/sericite + alunite + siderite + hematites, which becomes argillic, with quartz + kaolinite + alunite + illite/sericite. There is finally an external propylitic alteration in which feldspar and hornblende remnants are kept, and which is more regional. None of the underground mining works related to these mineralized veins operated below the 100 m of the current surface area, thus indicating that mineralisation was restricted to the zone close to the paleosurface. In the greatest depth, the mineralized structures become considerably narrower, and gold grades fall dramatically.

Deposits are temporally and spatially closely related to porphyritic intermediate-composition magma emplaced along precaldra structures but unrelated to the caldera forming magmatic system. The last phase of volcanic activity in the caldera complex was the emplacement of hornblende andesite flows and intrusions (Rytuba et al., 1990). This magmatic event resulted in structural doming of the caldera and opening of fractures and faults used as fluid channels, and provided the heat source for the large hydrothermal systems which deposited quartz-alunite type gold deposits and base metal vein system.

From a climatic point of view, this region is characterised by its dryness, showing a semi-desert kind of climate. Unusual and intense precipitations, together with scarce vegetation, result in a strong run-off with flooding and important land erosion. Regarding land cover, there is an abundance of bare soils with very dispersed and scarce vegetation. This scarce vegetation, together with its lithological/geological characteristics, make this area a favourable sector for remote sensing studies, as shown by diverse pilot studies carried out in the area in recent years (Bedini et al., 2008; Escribano et al., 2010; García et al., 2008).

4. Data and methods

4.1. Exploration criteria and data

Rigol-Sanchez et al. (2003) and Chica-Olmo et al. (2002) integrated all available information, facilitated by ADARO, S.A. and collected during DARSTIMEX Project (University of Granada), related to the synthesis of gold in Rodalquilar within a geodatabase, on the basis of the deposit model for the district outlined by Rytuba et al. (1990). The database is constituted by 46 gold occurrence locations, corresponding to exploited deposits and known mineralised structures and physical-chemical data such as a geochemical survey (59 elements, 372 locations), gravity and magnetic survey (330 ground stations) and geological information regarding fractures and lithology. Although in previous studies Landsat 5 Thematic Mapper (TM) images were used (Chica-Olmo et al., 2002; Rigol-Sanchez et al., 2003), in this paper the information provided by a hyperspectral EO1-Hyperion image is evaluated.

One of the main exploration criteria was finding the presence of a dioritic magma, the heat source for the large hydrothermal systems, at the base of the volcanic pile. Gravity and aeromagnetic data show a geophysical anomaly coincident with the alteration zone and reflect the presence of dioritic magma emplaced in the base of the volcanic pile. The structural control of mineralisations is evident in the light of the deposit model, therefore finding fractures and subsequently using them in different analyses was another key criteria. The existence of alteration zones was another important aspect to consider in the study of mineralisations in the sector. Hyperspectral remote sensing can be used

to recognise surface alteration mineralogy and alteration zones in regions where the bedrock is exposed. On the one hand, information on hidden altered zones can be provided by gravimetric and magnetic geophysical data in depth. Bouguer reduction density technique determined a lower average density of acid volcanic rocks ($2.20\text{--}2.30\text{ g/cm}^3$) than the reference density value (2.5 g/cm^3). This lower density may be partly due to changes in texture caused by the epithermal alteration to which the presence of Au is associated. In any case, these are two indirect prospecting methods based on the measurement of physical qualities of rocks/minerals (density and magnetic susceptibility) which have had a limited interest in the mineral prospectivity activities in the region, less than geochemical prospecting or remote sensing. The geophysical signal of these features is in many cases only an indirect evidence of mineralisation; in this case, in order to obtain information about geometries of subsoil materials such as lineaments. These lineaments can be interpreted as flow preferential directions of hydrothermal fluids. On the other hand, the geochemical signature of this type of deposits presents high Au + As + Cu values, increasing the presence of base metals and Te in depth (Cox and Singer, 1986). Moon and Evans (2006) also point out that Sb, Sn, Hg, Te, Se, S, and Cu are chemical elements associated with epithermal deposits of precious metals which serve as a basis for geochemical prospecting. Therefore, a priori, geochemical methods will attempt to basically detect the presence of the associated elements As, Ag, Sb, Cu, Sn, Te, Se, S, Fe, Pb, and Sr, apart from Au. Lithogeochemistry focused mainly on detecting hydrothermally altered rocks with high quartz or silica, kaolinite, alunite, pyrophyllite, illite/sericite, siderite, hematites and jarosite content. Attention was also paid to rocks with high K content (Si, Na and Ca).

The Rodalquilar Hyperion scene was acquired on 6th February 2006. Hyperion is a satellite-borne hyperspectral sensor which provides continuous spectral coverage of 242 bands, in approximately 10 nm sampling intervals, over the reflected spectrum from 0.4 to 2.5 μm , which makes it especially suitable for geological applications (Kruse et al., 2002). The instrument consists of two detectors. The VNIR (VIS + NIR) detector covers a spectral range of 400–1000 nm in 70 channels and the SWIR detector covers the range of 900–2500 nm in 172 channels. The spatial resolution of the image is equal to 30 m for all spectral bands. Pre-processing of the Hyperion image included removal of noisy and inactive bands (1–7, 57–77 and 225–242) (Beck, 2003) and destriping to correct the vertical striping patterns in the data. The Hyperion data were then converted to apparent reflectance using FLAASH (Berk and Adler-Golden, 2002; RSI, 2007). Hyperion data were used to derive spectral information that could be related to the alteration zones associated with gold mineralization in Rodalquilar. However, the direct mapping of either minerals or alteration zones was not possible due to the spatial resolution characteristics of the sensor. The method proposed by Boardman and Kruse (2011) was followed to carry out the unsupervised mapping of the image's different distinguishable spectral classes (note that these classes do not correspond to mineral or alteration classes). A Minimum Noise Fraction (MNF) procedure was used to reduce the residual noise after the destriping process. MNF is an orthogonal transformation which orders the images by the signal-to-noise ratio (Green et al., 1988). The process of endmember selection continued by keeping the first 7 bands of MNF transforms, which contained most of the spectral information (see Fig. 2). The dimensionality of the transformed hyperspectral data was determined by comparing both the eigenvalue plots and the MNF images. A PPI (Pixel Purity Index) (Boardman et al., 1995) algorithm was applied to locate the most spectrally extreme pixels. The PPI was computed by repeatedly projecting the 7-dimensional scatterplots onto a random unit vector and recording the number of times each pixel was marked as extreme. Finally the Mixture Tuned Matched Filtering (MTMF) algorithm was used (Boardman and Kruse, 2011) in order to map the abundance of the endmembers selected. MTMF maximised the response of the known endmembers and suppressed the response of the composite unknown background. It is worth mentioning that other algorithms such

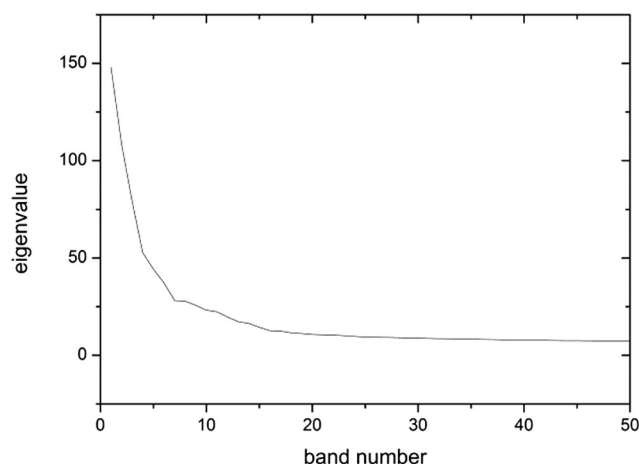


Fig. 2. MNF eigen value plot. Higher eigen values generally indicate higher information content.

as Spectral Angle Mapper, or Linear Spectral Unmixing obtained worse results than the previous case. The procedure described in Bedini et al. (2009) was used to identify the minerals compressed by endmembers. The MTMF bands representing different spectral classes were used as inputs to the model, without being able to find a clear correspondence between the endmembers automatically selected and the spectra of different mineral species could not be found.

Geochemical data were processed by performing principal component (PC) analysis on 46 selected mineralisation-related elements. Gold is usually found in association with areas affected by silicification, or in zones where processes of alunite or jarositic alteration have occurred. Sulphur is closely related to hydrothermal alteration processes in the presence of jarosite or alunite-type sulphates and is therefore associated to mineralisation of gold. PC1, PC2 and PC3, indicating lithology, were chosen for further modelling. PC1 showed the presence of SiO_2 and Al_2O_3 and the absence of CaO; PC2 was composed by Pb, Zn, Cu and W; and, PC3 by As, S, Ag Au and Th. Continuous layers were created by kriging (Chica-Olmo et al., 2002) from the PC scores to minimise estimation error. Gravity and magnetic residual values were also interpolated by kriging to generate residual anomaly maps indicating the presence of potential ore-related buried anomalous bodies. A map of distance-to-nearest-fracture map was derived from the fracture map by using GIS analysis functions. The mineralised structures are related to N–S regional fractures, and to ring and radial fractures associated to caldera margins which generated permeable zones. The deposits are located in vertical veins and fractures in silica-rich rocks, in silicified hydrothermal breccias and in chalcedony which fills fractures and cavities. The lithology map of the area was reclassified into 4 classes: very favourable, favourable, less favourable and non-favourable (Table 1). It should be noted that this last layer was incorporated to the feature vectors as a categorical feature (i.e. A, B, ...) (see Section 5.1 of Breiman (2001)). The deposits are linked to ignimbrite dacites and rhyolites, with a high K content and intensely altered generally, with

Table 1
Description of lithological gold favourability categories.

Class Id.	Category	Description of the category
1	Very favourable	Pyroclastic and ignimbritic flows, reddish-purple biotite-amphibole dacite and ignimbritic dacites with tuffs and basal ignimbrites.
2	Favourable	Dacite–rhyolite tuffs and pyroxene andesite.
3	Little favourable	Fine grain quartz-anfibolitic dacite domes and flows, pyroclastic breccias and ash-flow tuffs of anfibol, amphibole andesite and dacite.
4	Non-favourable	Calcareous sediments; alluvium/colluvium and andesite breccia.

approximate ages ranging from 12 to 9 Ma. The deposits are located in vertical veins and fractures in silica-rich rocks, in silicified hydrothermal breccias and in chaledony which fills fractures and cavities. Wall-rocks are mainly tuff, ignimbrites, breccias and rhyolite domes (Arribas et al., 1995).

The thematic layers in the Rodalquilar database were combined into a set of input feature vectors at each cell location in the set of grids. These vectors formed the input to the MLA algorithms and are known as input-feature vectors. Known deposit locations were used as a response feature for the training of the algorithms. Training patterns were created by recording the input feature vector values at each of the 46 locations of the gold occurrence database. The training dataset was completed adding 57 sterile locations scattered over the district selected by means of stratified random sampling within little or non-favourable lithological locations which were distal to existing gold deposits. Each training pattern consisted of an input feature vector paired together with a binary target value (target values used in the training data were 1 for gold occurrences and 0 for non-gold occurrences). Hence, the output of the algorithm will be a floating value ranging from 0 to 1, representing the probability of mineral deposits.

4.2. Induction of MLA models

Data processing for the induction of the MLA consisted of three main stages: (i) training and parameterisation of the algorithms; (ii) post-processing requiring converting the output values to a map; and (iii) accuracy assessment. All of the MLA models were created using the R 2.10.1 (R-Project) free software. Within this environment, “rpart” libraries were used for inducing decision trees, “nnet” for feed-forward neural networks, “e1071” for support vector machines and “randomForest” for random forest.

In order to study the performance of the different machine learning algorithms it is very important to determine a suitable combination of parameters, which allows generating operative robust predictive models, avoiding the application of the default settings recommended by the commercial software used. Additionally, studies which assess a new algorithm, comparing it with other methods, are likely to be biased as a consequence of a better knowledge of the studied method (Mas and Flores, 2008). In other words, the parameterisation of the proposed algorithm becomes optimal, while a greater uncertainty exists in the parameterisation of the rest of algorithms. On the contrary, if no substantial differences in the accuracy of the methods exist, the comparison among algorithms should be based on other factors such as operational capacity, ease of use or the interpretability of results.

4.3. Validation of predictive models

To assess the optimal value of the different parameters of every method, the predictions derived from all possible parameter combinations were evaluated using the Mean Square Error (MSE) using a 10-fold cross validation procedure. The “best” model was the one with the lowest MSE. The methodology followed in the selection of optimal parameters of each method was based on a manual search for them, since one of the goals of this study is to show variation in the mapping accuracy of results according to the parameter selection. In the context of machine learning, other methodologies exist to solve problems related to model selection/parameter optimisation such as grid search, genetic algorithms or random search, which can be used to automatize this process (Bazi and Melgani, 2006; Bergstra and Bengio, 2012). The best-fit models resulting from the application of each of the methods were compared in terms of success rate and ROC curves (using training data points as a validation reference). The success rate was computed reclassifying the gold potential maps according to different thresholds of areal percentages of prospective zones and calculating the success rate of those prospective zones against the known gold occurrences (true positive rate; TPR) (Agterberg and Bonham-Carter, 2005). The

success rate is the percentage of training deposits delineated correctly in prospective zones. In this study, reaching a high success rate for the smallest possible prospective area is essential, given that the exploitation costs are directly related to the extent of the prospective area. Model performance curves were then created by plotting percentages of prospective zones versus success rates. However, in this analysis using the success rate, the false positive rates (FPRs) are ignored. Therefore, an analysis which considers both types of rates (TPR and FPR) was carried out through the calculation of ROC curves, in which the prospectivity area can be controlled by means of the FPR, i.e., the proportion of bare pixels considered as mineralised. ROC curves were plotted by varying the threshold on the predicted output. The ROC curve gives a graphical representation of these TPR and FPR for various thresholds on the output. A threshold will determine if there exists gold or not. If the likelihood was greater than the threshold, the predicted class would be 1 or “gold occurrence” and if lesser than the threshold, the predicted class would be 0 or “non gold occurrence”. Generally, the false positive rate (FPR) result is plotted on the x-axis vs. the true positive rate (TPR), which is plotted on the y-axis. Each threshold results in a (TPR, FPR) pair and a series of such pairs are used to plot the ROC curve. These are also known as the “sensitivity (TPR)” and “specificity (1-FPR)”. The area under the curve statistic (AUC) was used to determine which models performed better. An AUC value of 1 is considered perfect and an AUC value equal to 0.5 is considered as random guessing (Bradley, 1997).

In the modelling of many real-world exploration scenarios the availability of training data is limited. However, it is necessary that the number of training areas be large enough to represent all the variability of the mineral deposits under study, in order to reach an acceptable mapping accuracy level. Additionally, for certain mining districts the availability of data is limited. The effect of the training set size on MLA performance was evaluated using the Kappa index of accuracy, reducing the training sets in increments of 10%.

4.3.1. Artificial neural networks

Different factors affect the capacity of ANN to generalise, i.e., to predict new data from the learning carried out with training data. The intrinsic factors to network design include: number of neurons and network architecture. The problem of how to define the most suitable network architecture is related to the nature of the hidden layer. There is no rule for determining the number of hidden layers, but, theoretically, one single hidden layer can represent any Boolean function (Atkinson and Tatnall, 1997). In general terms, the higher the number of units of the hidden layer, the greater the network capacity to represent the training data patterns. However, the fact that the hidden layer has a high number of units also produces a loss in the networks' generalisation power (Atkinson and Tatnall, 1997; Foody and Arora, 1997).

Numerous supervised standard feed-forward propagation neural network models were built using a standard sigmoid transfer function. To this end, neural networks of different architectures were trained, made up of a single hidden layer, whose number of units was set between 1 and 10. Likewise in order to optimise the network training, the range of initial weights assigned by the network was set between the interval 0 to 1, with increases of 0.05. From these initial values, different weight decay values were considered (between 0.01 and 0.1 at 0.05 intervals). The optimal value of weights was set by means of least squares.

4.3.2. Regression trees

It is necessary to set a series of parameters for the training of decision trees, such as dissimilarity measure, the depth of the tree and the minimum number of observations per node. The dissimilarity measure or heterogeneity influences the way in which the algorithm performs data splits in each node. The depth of the tree and the minimum number of observations are parameters linked to the structural complexity of trees: the more the number of levels and the less the number of minimum observations in nodes, the greater the structural complexity of

the model. Hence, it is necessary to set these parameters in order to achieve the highest accuracy in the prediction, avoiding the creation of complex tree structures which over fit data and lose generality (Pal and Mather, 2003). For this study, CART decision-tree models were used (Breiman, 1984). For the induction of trees, the Gini index was considered as a dissimilarity measure (Breiman, 1984; Quinlan, 1993). With the aim of obtaining robust and generalizable models, all possible decision-trees were assessed, for depths of tree from 2 to 29, with a minimum number of observations per node between 1 and 50.

4.3.3. Random forest

Unlike most methods based on machine learning, RF only needs two parameters to be set for generating a prediction model: the number of regression trees and the number of evidential features (m) which are used in each node to make regression trees grow (Rodriguez-Galiano et al., 2012b). Breiman (1996) demonstrated that by increasing the number of trees the generalisation error always converges; hence, overtraining is not a problem. On the other hand, reducing the number of m brings as a result a reduction in the correlation among trees, which increases the model's accuracy. In order to optimise these parameters, a large number of experiments were carried out using different numbers of trees and split evidential features. The range of the number of trees was set between 1 and 1000 at intervals of 2, and the number of splits evidential features, between 1 and 15, at 1 intervals.

4.3.4. Support vector machines

SVMs need the adjustment of a high number of parameters for their optimisation: a) Linear, polynomial, sigmoid and radial basis (RBF) kernel functions; b) cost; c) gamma of the kernel function, with the exception of the linear kernel; d) bias on the kernel function, only applicable to the polynomial and sigmoid kernels and, finally, e) degree of the polynomial, only applicable to the polynomial kernel. The adequate value of these parameters is data specific, therefore it is necessary to optimise them in order to get generalizable models; i.e. these must not over fit or under fit data, therefore they must be accurate (Abedi et al., 2012; Cortes and Vapnik, 1995; Yang, 2011; Zuo and Carranza, 2011).

We used SVM of RBF as it was reported by Zuo and Carranza (2011) that the errors for RBF and polynomial kernel were lower compared to linear and sigmoid kernels. However, RBF has less parameters to tune, as there is not polynomial degree parameter. In order to assess the impact on the mapping accuracy of each of the abovementioned parameters, a set of SVMs were built for different parameter combinations. For the building of SVM, the cost was fixed between 0.1 and 50, at 0.1 intervals; gamma between 0.05 and 1, at 0.05 intervals.

5. Results and discussion

5.1. Sensitivity of MLAs to parameter configuration

The parametrisation of MLA has a great influence on their robustness and generalisation capacity, and hence in the accuracy to predict new gold occurrences. Fig. 3 and Table 2 show significant differences in the accuracy obtained by the different machine learning methods according to the parameter setting used. SVM models were less accurate than the rest of the methods, reaching the highest average MSE errors (mean of 0.19, standard deviation of 0.03). However, RF was very robust and stable, with the lowest average and standard deviation MSE values (mean of 0.12, standard deviation of 0.01). Fig. 3 shows that all MLA methods (with the exception of RF) are very sensitive to variations in the parameters used for their training; the optimal error values reachable by each algorithm take place for very determined parameter combinations, especially for the case of ANN. This confirms the results by Rodriguez-Galiano and Chica-Rivas (2012), who in a study about land cover mapping found that ANNs present a greater sensitivity than the rest of algorithms. However, RF apart from being an operative method in terms of the simplicity of its parameters, also presented a greater

stability against variations in its internal configuration (see Fig. 3 and standard deviation in Table 2). This better performance of RF can be attributed to the combination of multiple individual classifiers, trained under very particular conditions. On the one hand, the fact that the evidential features used for the induction of each tree are chosen randomly reduces the correlation between individual models, which reduces the generalisation error and provides predictions with great stability. Although regression trees in isolation are less robust than a regression tree trained using the best evidential features for splitting in each node, the set of trees (average) is more accurate. Additionally, to the way features are selected must be added the resampling of training data for each tree (bagging), which also contributes to increasing the diversity of models which make up the ensemble and prevents trees from over fitting the data. Below mapping accuracy is quantitatively analysed with relation to the different parameters used in the building of each type of classifier.

The RT models with the best performances were created by using the Gini index as a measure of heterogeneity, between 29 and 31 minimum numbers of samples in every node. The maximum depth of the tree did not affect results. The error was significantly higher when nodes of less than 20 samples were allowed, which means rules were created to split a small number of samples. Hence, it is preferable to limit the number of samples in terminal nodes so that these do not over fit the data and, hence, the model does not lose generality in turn (Pal and Mather, 2003).

RF incorporates an additional parameter which is not considered in traditional decision trees: the m parameter. This m value remains constant while the tree grows, and the selection of evidential features is random. From about 50 trees the Kappa value converged up to an MSE of 0.11 for m between 1 and 6. The addition of more trees neither increased nor decreased the generalisation error. However, an important increase in computation time was observed when a high number of trees was considered. Ensembles made up of few regression trees produced poor results, while greater ensembles produced more accurate prospectivity models.

Regarding ANN the architecture has a significant impact on its ability to predict mineral potential correctly. Generally, the largest and most complex networks are more effective in order to define a training dataset. However, these types of networks perform worse generalisations than smaller and simpler networks. The mapping accuracy increased as the network became more complex, i.e., it increased with the number of units of the hidden layer. The minimum error was obtained for neural networks with a number of units in the hidden layer equal to 6, 7 or 9, for very specific weight decays.

The training of SVM was also complex; the parameters involved in the optimisation of the RBF kernel function were assessed individually. From this initial evaluation, it was possible to build the optimal model on which the comparison was based. Fig. 3 shows how the cost parameter had a limited effect on the model's accuracy. For cost values greater than 1 the error converged in most cases, with the exception of gamma values lower than 0.1. As cost grows, and a greater number of errors is allowed, the model's accuracy increases until reaching a balance between the number of errors allowed and the model's generalisation power (Cortes and Vapnik, 1995). On the other hand, the gamma parameter strongly influenced the performance of the algorithm. This contrasts with the results of Zuo and Carranza (2011) who concluded that the accuracy of the model (in this case classification model) was not sensitive to the choice of gamma. It should be noted that in the cited work gamma varied between 0.25 and 1000, therefore the sensitivity of this parameter, usually fitted to small values, could be masked. Minimum error values were obtained for costs over 1 and gamma values in the range between 0.15 and 0.2, which indicates that the training data used in the calibration of the algorithms had a very low number of outliers. This parameter, gamma, is traditionally fixed to the value of the inverse of the number of input features, 0.067 in this case (Yang, 2011). However, in view of our results, we believe the joint adjustment of both parameters, cost and gamma, to be more suitable.

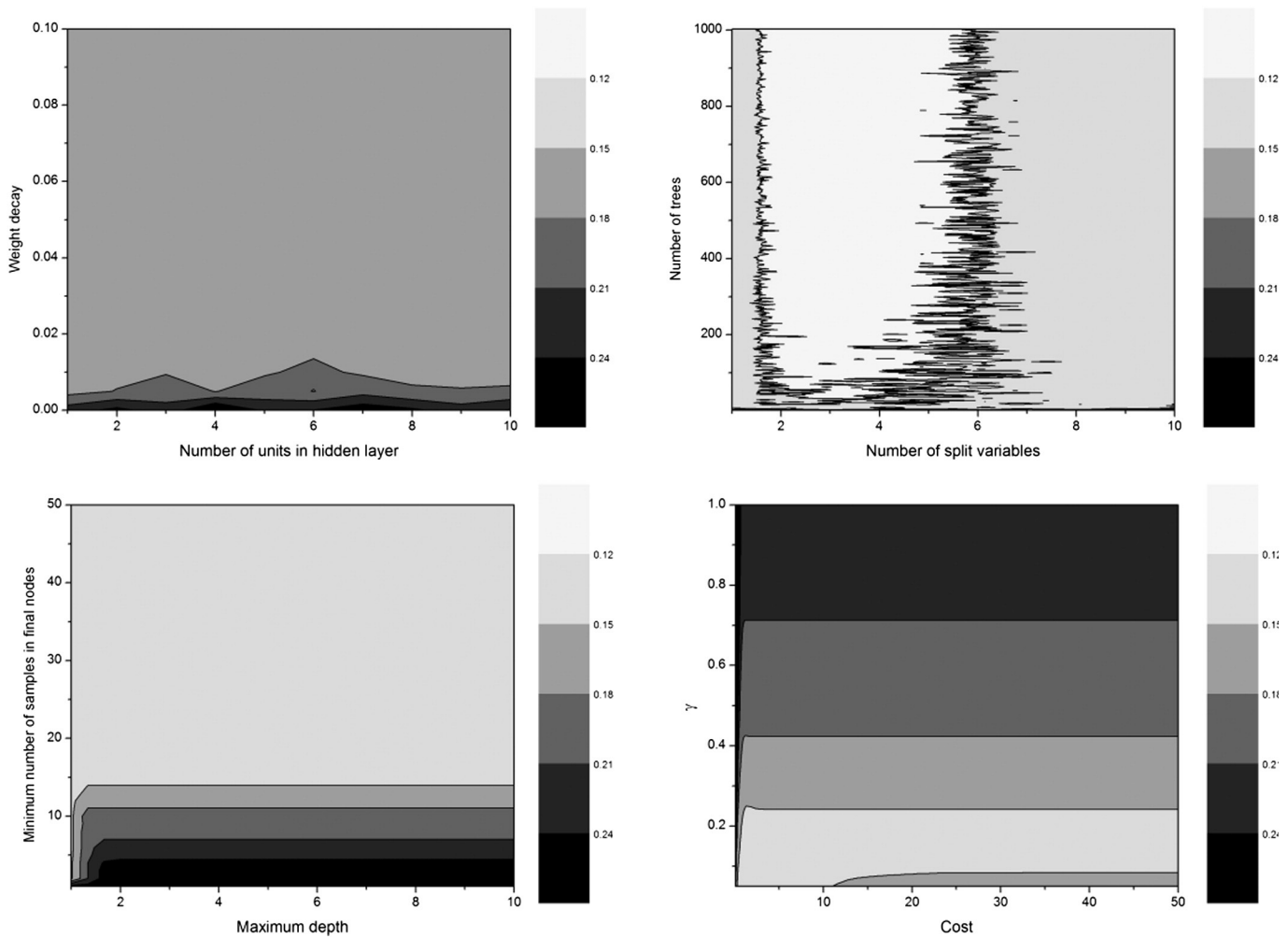


Fig. 3. Mapping accuracy (MSE) for all the parameter combinations used in the training of every MLA method.

5.2. Accuracy of gold potential models

Gold potential maps produced using the MLA methods trained using the optimal parameter configurations are shown together with gold occurrence points used in the training in Fig. 4. Areas with higher gold potential are located mainly in the central part of the study area and around fracturing and faults identified as highly prospective areas (see Section 4.1). It can be seen how there is a high correspondence between the deposit area delineated by each method and the information obtained from the Hyperion image (see the false-colour composite shown in Fig. 3). From a visual point of view, it can be observed as ANN assigned higher values to deposit areas located to the East of the study area, while RF and SVM, distinguished between a deposit main central core and marginal areas with a smaller probability. It is worth pointing out that RT was only capable of assigning four different occurrence probability values: low probability (0.023), medium–low (0.462), medium–high

(0.5) and high (0.944). In this latter case, some Au deposit evidence appears in medium–low probability areas.

Because in MLA regression modelling the predictions are floating values ranging from 0 to 1 denoting the likelihood of mineral deposit occurrence, output values of ≤ 0.5 are classified as non-deposit and values of > 0.5 are classified as deposit (see right column in Fig. 4). However a more rigorous reclassification of probability maps can be carried out using a ROC analysis (see Section 4.2 and the last part of the current section). Considering this reclassification of the output maps, Table 3 shows that RF outperformed the rest of the methods with Kappa and overall accuracy values equal to 0.92 and 0.96, respectively. SVM also had a good performance with Kappa and overall accuracy values that can be considered as very satisfactory (0.87 and 0.93, respectively). On the other hand, ANN, and specially RT, brought about less accurate mineral prospectivity maps, with Kappa values equal to 0.77 and 0.66 and overall accuracy values equal to 0.89 and 0.83, respectively. These results confirm what other authors have identified in different modelling problems using satellite images for the classification of land covers: ANN and RT have a tendency to over fit data and lose generalisation power. From the standpoint of differentiating between deposit and non-deposit areas, RF also achieved better results, being able to delineate both areas in a balanced way (Kappa equal to 0.92 for both categories). In the case of ANN and SVM, non-deposit areas were more accurately delineated, which can be contradictory, given that the reliability of deposit locations is possibly greater than that of non-deposit ones, as the former are identified on the basis of objective evidence. However, this effect could be related to the number of examples used

Table 2
Accuracy of MLA modelling obtained from all the hyper-parameters combinations.

	ANN	RF	RT	SVM
Min	0.16	0.11	0.13	0.13
Max	0.28	0.31	0.27	0.37
Avg	0.17	0.12	0.16	0.19
St. dev.	0.02	0.01	0.04	0.03

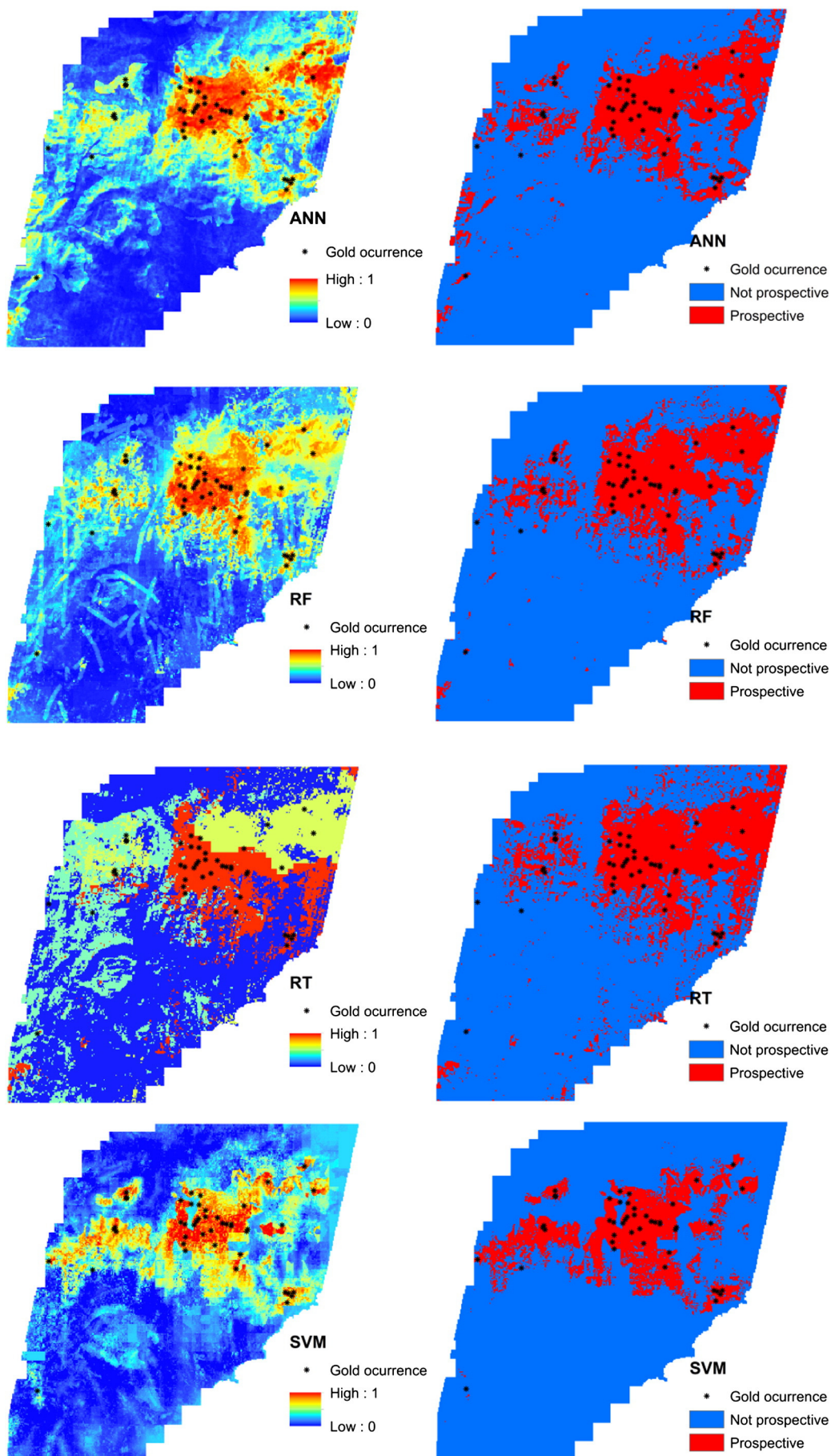


Table 3

Accuracy of the best model obtained for every machine learning method.

	ANN	RF	RT	SVM
Overall accuracy	0.89	0.96	0.83	0.93
Kappa	0.77	0.92	0.66	0.87
Kappa deposits	0.74	0.92	0.66	0.82
Kappa non-deposits	0.80	0.92	0.66	0.92

in the training (46 deposit locations against 57 non-deposit ones), therefore these algorithms tended to bias the prediction to maximise the accuracy of the largest class. Regarding, the percentage of deposits classified as prospective areas RF showed the better performance (97.83%), followed by SVM (91.30%), ANN (86.96%) and RT (82.61%). Further experiments varying the ratio between deposit and non deposit locations should be explored.

To further evaluate the performance of the predictive maps obtained for the optimum training of every MLA algorithm, the success rate curves described in Section 4.2 were represented. Fig. 5a shows the success rate in the estimation of the known gold deposits according to different percentages of prospective areas. The area defined as highly prospective is significantly smaller in the RF map compared to the rest of MLA models. Hence, in order to reach a success rate similar to RF, other MLA methods need to delimit larger prospective areas. It can be observed how RF and SVM start from a similar success rate, although the slope of the success rate curve is steeper in the case of RF. The success rate of RF and SVM is over 90% for percentage threshold values of prospective areas over 10%, whereas for ANN is only equal to 70%. However, RF success rate converged at a 98% success rate value when 15% of the study area was considered as prospective. SVM needed to delineate 35% of the area to reach this success rate value. RT experienced the worst success rate reaching values over 95% only for areas greater than 75%. The success rate according to different percentages of affected areas only accounts for true positive rates (TPR), while false positive rate (FPR) is ignored. Fig. 5b shows the results of a ROC analysis which considers both TPR and FPR according to different probability threshold values of mineral prospectivity. It can be observed a very good performance of RF and SVM with very similar AUC values (0.999 and 0.998, respectively). ANN and RT were less accurate than the rest of the models, with AUC values of 0.962 and 0.907.

Fig. 6 shows the sensitivity of MLA models to the training set size reduction. It can be seen how a decrease in the accuracy takes place which is initially greater (data reduction of 10%). Generally all methods react in a gradual way to training data reduction. However, ANN and RT present less stable behaviours for certain reduction thresholds. For a reduction of the training set of 50% all the methods with the exception of RT obtained Kappa values higher than 0.70 (ANN: 0.70, RF: 0.73 and SVM: 0.71). From the 70% reduction threshold (18 positive occurrences), accuracy decreased more abruptly for all the methods. From Fig. 6 it can be observed how differences among methods grow as reduction increases. Hence, when only 6 positive occurrences were used (90% reduction), the map generated from RF presents a Kappa value of around 0.6, while the accuracy of SVM and ANN maps was equal to 0.52 and 0.47, respectively. In the case of RT, the generated model was completely inaccurate, with a Kappa value of 0.03.

5.3. Interpretability and transparency of models

As seen from Fig. 4, all maps present a similar distribution broadly speaking, although the probability values assigned can vary greatly among methods. These similarities and differences, as well as the mathematical bases of each applied algorithm, are due to the use each method gives to the evidential features. There are different ways of

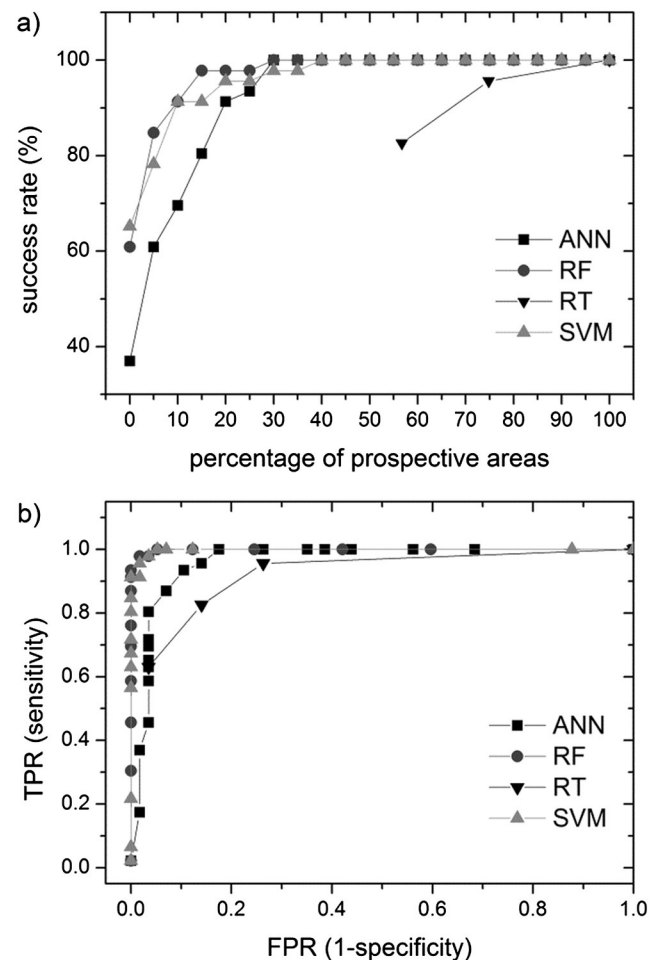


Fig. 5. Success rate (a) and ROC curves (b) of MLA predictive maps of epithermal gold prospectivity obtained using the best parameter configuration.

evaluating the importance of evidential features in a model (Guyon and Elisseeff, 2003; Rodríguez-Galiano et al., 2012a). On the one hand, distant approximations to the method (wrappers) can be used, such as not using some features and calculate the difference between the accuracy achieved by the model which used all the features and the models resulting from excluding each of them. On the other hand, there are modelling methods which integrate an approximation for the calculation of the importance of features (embedded). It is the case of RT and RF. However, both ANN and SVM are black-box techniques, and do not provide information about the role of features in the predictive modelling.

RT, although not as robust as the rest of algorithms (see Section 5.2), is the one to provide more information about how evidential features behave with relation to mineral deposits. Fig. 7 shows a tree diagram from which the rules used for the splits performed in each node can be deduced. The MTMF5 component, obtained from the Hyperion hyperspectral image, was the most informative evidential feature, as it allowed distinguishing between low and high deposit probability areas. Furthermore, low deposit probability areas can be subdivided into very low probability areas (0.023) and medium-low probability areas (0.462) on the basis of a higher or lower distance to fractures. Very high probability areas (0.944) and medium-high probability (0.5) where distinguished on the basis of the MTMF4 transform component of the Hyperion image. This method also provides the threshold values of the evidential features for which the subsplit takes place,

Fig. 4. Predictive maps of likelihood values of epithermal gold prospectivity obtained for all MLA methods (left panel) and reclassified gold potential maps considering a likelihood threshold value of 0.5 (right panel).

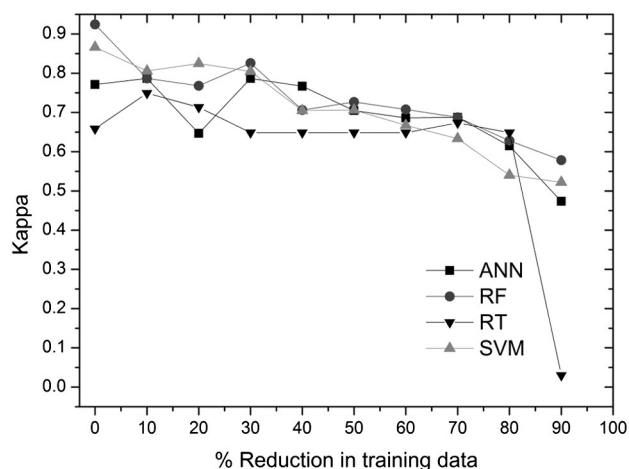


Fig. 6. Effect of reducing training data on the mapping accuracy of MLA predictive models.

although in this case their interpretation in absolute terms is limited, given the features were normalised.

RF also allows estimating the importance of each evidential feature in the model, although, unlike an RT model, it does not allow to identify threshold values in the evidential features. Fig. 8 shows the result of an internal calculation carried out by the algorithm in which the difference in the MSE is calculated, which is a consequence of not using each of the features. As in the case of RT, RF identified MTMF5 as the most important feature, assigning it much greater importance than to the rest of features. Therefore, although it was not possible to confirm the correspondence between endmembers and mineral species, we believe that the spectral information contained in the MTMF5 component may be related to high alteration zones linked to Au mineralisation in Rodalquilar. However, this claim can be regarded as speculation and, therefore, must be dealt with in future papers, perhaps comparing the spectral information derived from Hyperion to other higher resolution hyperspectral sensors (i.e. HyMap) for the mapping of Au potentiality using random forest.

Also distance to fractures and main component number 3 of geochemistry were of a significant importance in the model (see Section 4.1). Fracture zones are important for modelling as they provide active pathways as well as physical traps for gold-bearing fluids responsible for Au epithermal mineralisation in this area. In turn, main component number

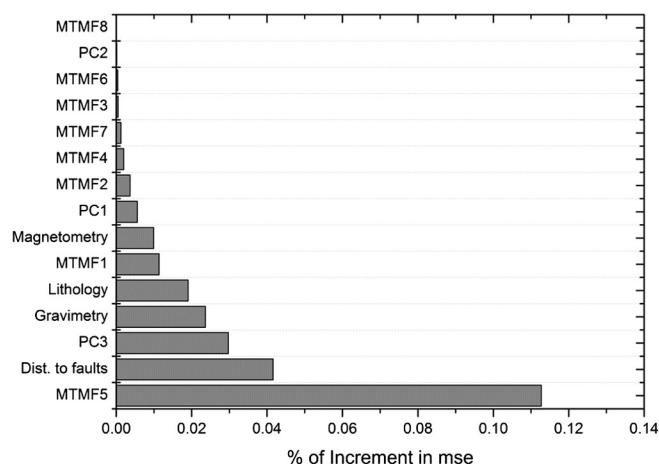


Fig. 8. Importance of predictive variable in RF prospective model.

3 of geochemical data corresponds to volcanic rocks linked to high fracturing and hydrothermal alteration areas.

Rodriguez-Galiano et al. (2014) in the study in which RF was presented for mineral potential modelling also estimated the importance of evidential features in this district. However, the spectral features (satellite data) used for the modelling were different. In the first case Landsat images were used, while in the present study Hyperion images have been used, therefore with a much greater amount of spectral information. The suitability of using an image with a greater spectral resolution is clear in that in the present study a component obtained from satellite images is significantly more important than the rest of the features, while when multispectral images were used, satellite data were of a minor importance than geochemistry or distance to fractures.

The RF model which used the Hyperion data estimated a greater prospective area than that estimated by a RF which used multispectral data from the Landsat sensor (21.53% and 16.65%, respectively) (see Fig. 9). The percentage of deposits classified as prospective areas were equal to 97.83 and 95.65 for Hyperion and Landsat models, respectively.

6. Conclusions

The comparative analysis of the MLA methods for modelling mineral prospectivity was carried out from different perspectives: ease of application and effectiveness, sensitivity to the configuration of the model's

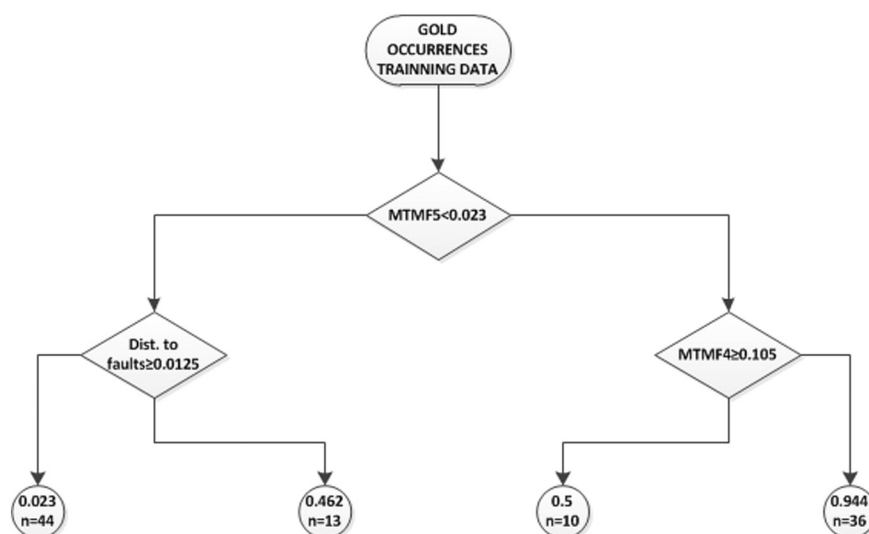


Fig. 7. Scheme of RT prospective model.

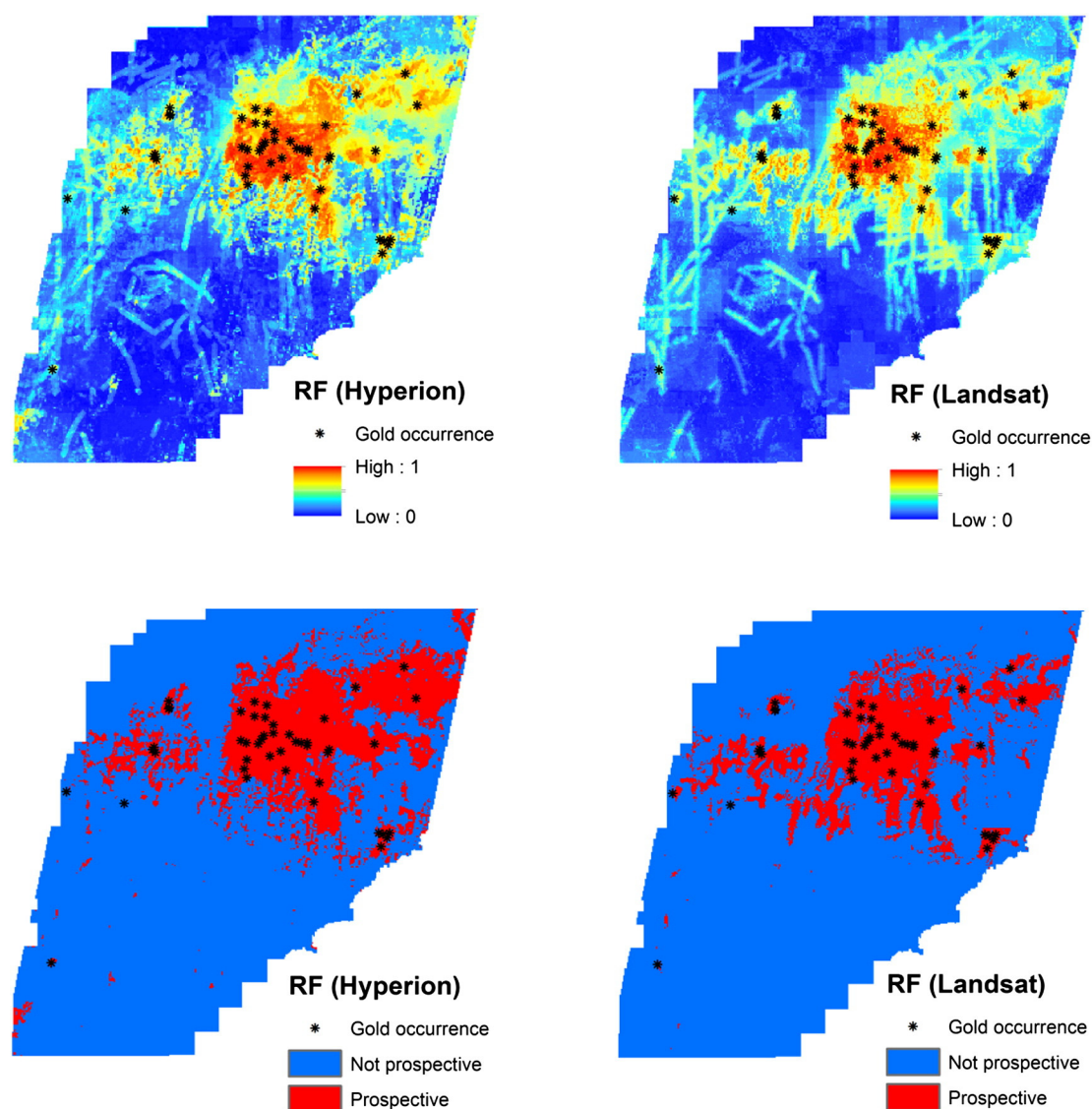


Fig. 9. Predictive maps of likelihood values of epithermal gold prospectivity obtained for RF using Hyperion or Landsat data (top panel) and reclassified gold potential maps considering a likelihood threshold value of 0.5 (bottom panel).

parameters and data reduction, the mapping accuracy of classifications, and transparency and interpretability of the models.

The assessed models have a different difficulty in their training. Decision-tree-based algorithms (RT and RF) involve a lesser difficulty in their training. This applies to both simple regression trees and ensembles of trees (RF). However, ANN and SVM are more complex. SVMs are based on different kernel types, according to which the combination of parameters to be optimised is different.

The greatest accuracy of classifications was achieved by RF and SVM, with Kappa values equal to 0.92 and 0.87, respectively. ANN also achieved an acceptable level of mapping accuracy (Kappa equal to 0.77), although only for a very specific combination of their adjustment parameters. Lastly, the maximum Kappa index derived from the RT model was considerably lower than that of the rest of methods (0.66). It is worth mentioning that this conclusion can only be applied to the best classification methods obtained from a complex optimisation process, since, in general terms, the performance of RF for all the parameter combinations was better than that of the rest in terms of stability and accuracy. Regarding the results of classifications per categories, the choice of method resulted in differences in the accuracy of classifications according to positive or negative occurrences. RF managed to delineate both areas with equal accuracy, while ANN and SVM distinguished both

areas in a biased way, overestimating non-deposit areas. The rest of statistical measures used to compare map quality also indicate that the RF method performs better than the rest. The MSE, success rate and AUC values were higher for RF. However, it should be highly emphasised that no broader generalisations can be made about the superiority of any method for all types of problems. The performance of the methods might vary for other datasets. However, the outlook for the use of RF in mineral potential modelling research and applications is very promising.

The assessed algorithms responded in a similar way to the reduction of the number of training areas. However, when the data are very scarce RF showed a better performance being able to reach a Kappa index equal to 0.6 when only 6 deposit locations were used to train the model.

The RT and RF methods could estimate the importance of every single evidential feature in the modelling of mineral potential. Both methods found the information taken from the Hyperion hyperspectral image as key in the modelling of Au potential in this area.

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